

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 7

PHYSIOGRAPHIC DIVISIONS **THE PENINSULAR PLATEAU**

INDIAN PHYSIOGRAPHY

HILL RANGES OF THE PENINSULAR PLATEAU



- 1 Most of the hills in Peninsular region are of the **relict type**(residual hills).
- 2 They are the **remnant of the hills** formed many million years ago.
- 3 The Plateaus of the Peninsular region **are separated from one another by these hill ranges and various river valleys**.

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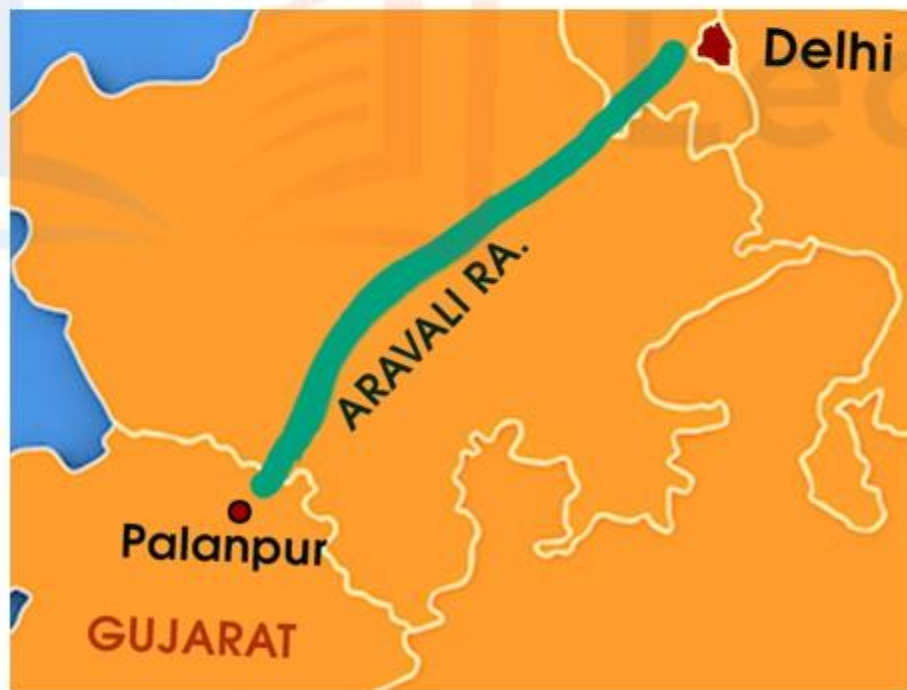
HILL RANGES OF THE PENINSULAR PLATEAU

1

ARAVALLI RANGES

1

One of the major physiographic elements of the Peninsular India is the Aravalli ranges.



2

This runs in a northeast to the southwest direction for 800 km between Delhi and Palanpur near Ahmedabad in Gujarat.

3

They are the one of the oldest fold mountains of the world.

4

Now, these ranges have become relict because of severe erosion that is happened for millions of years.

5

The range is very conspicuous in Rajasthan and becomes less distinct in Haryana and Delhi.



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HILL RANGES OF THE PENINSULAR PLATEAU

2

THE VINDHYAN RANGE



An escarpment is a steep slope that forms as an effect of faulting or erosion, separating two relatively level areas.

1 The Vindhyan ranges overlooking the Narmada valley rises as escarpment flanking the northern edge of the Narmada river valley.

2 It runs more or less parallel to the Narmada valley in an east-west direction.

3 The general elevation of the Vindhyan ranges is 300-650 m.



VINDHYA RA.

KAIMUR
HILLS

Narmada R.

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HILL RANGES OF THE PENINSULAR PLATEAU

2

THE VINDHYAN RANGE



4 Most parts of the Vindhyan range are composed of **horizontally bedded sedimentary rocks** of ancient age.

5 The western part of this range is covered with **lava**.

6 The Vindhyas are continued to eastwards as the **Kaimur hills**.

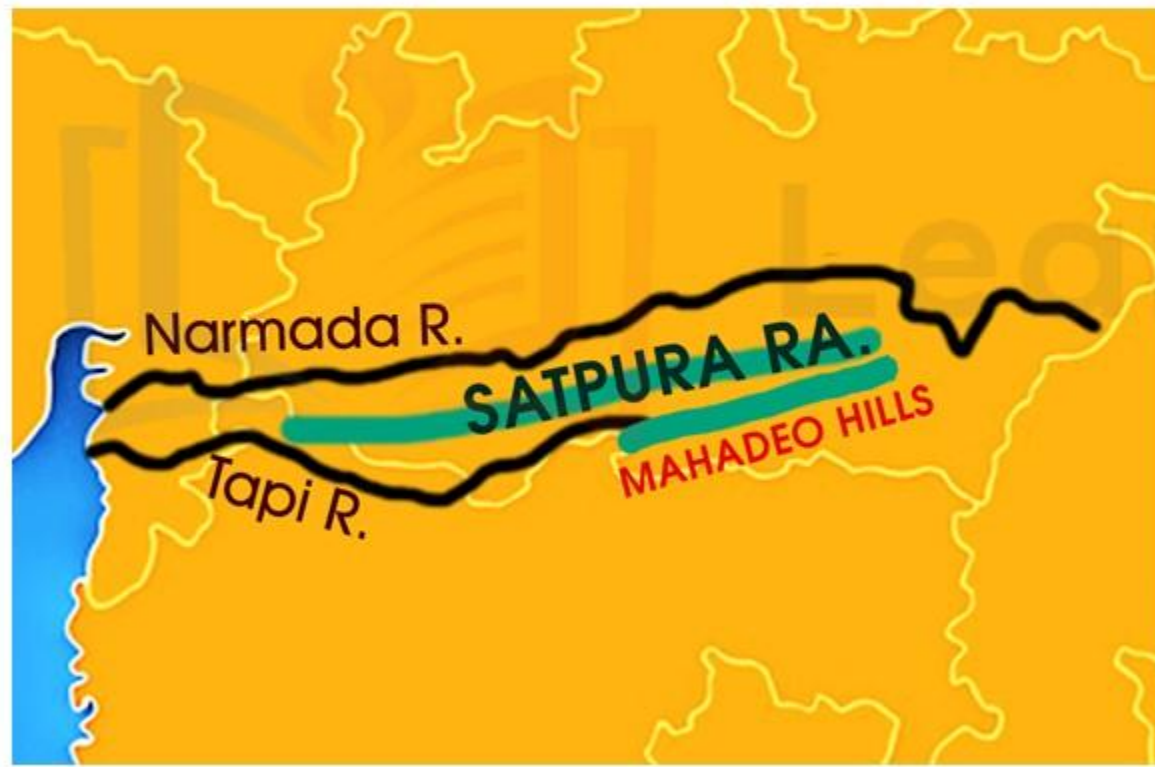
7 This **range acts as the water shed** between the Ganga river system and the river systems of South India.

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HILL RANGES OF THE PENINSULAR PLATEAU

3

THE SATPURA RANGES



- 1 Satpura is the **series of seven mountains**, 'Sat' in Sanskrit means seven and 'Pura' means mountains.
- 2 It runs in **east-west direction**, south of the **Vindhyas** and in between the **Narmada** and the **Tapi**, roughly parallel to these rivers.
- 3 It stretches for a distance of about 900 km. **Dhupgarh near Pachmarhi** on Mahadev Hills is the highest peak.



A map of Western India with a yellow background and white outlines of state boundaries. A thick black line represents the Satpura Range, running horizontally across the center. Below it, a thinner black line represents the Mahadeo Hills. The area between these two ranges is highlighted in light blue. The Narmada River is shown as a blue line on the left, flowing south. The Tapi River is shown as a blue line below the Satpura Range, also flowing south. The Arabian Sea is visible on the far left, and the Bay of Bengal is visible on the bottom right.

Narmada R.

SATPURA RA.

MAHADEO HILLS

Tapi R.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



- 1 This forms the western edge of Deccan plateau.
- 2 They run in north-south direction, parallel and close to the Arabian sea coast.
- 3 It runs from the Tapi valley to a little north of Kanyakumari for a distance of about 1,600 km.
- 4 The Western Ghats are steep-sided, terraced, flat-topped hills presenting stepped topography facing the Arabian Sea Coast.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



Stair case aspect of the Western Ghats

5 This is due to the horizontally bedded lavas, which on weathering, have given a characteristic stair aspect to the relief of this mountain chain.

6 The Western Ghats abruptly rise as a sheer wall to an average elevation of 1,000 m from the Western Coastal Plain and therefore they appear to be an imposing mountain.

7 But they slope gently on eastern flank and hardly appear to be a mountain when viewed from the Deccan tableland.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



Scenic Bhor Ghat in Maharashtra

- 8 The northern section of the Ghats from Tapi valley to a little north of Goa is made of horizontal sheets of Deccan lava.
- 9 Thal ghat and Bhor ghat are some of the important passes which provide passage by road and rail between the Konkan Plains in the west and the Deccan Plateau in the east.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



Train through Thal Ghat pass



*Train descending the last leg of
Bhor Ghat*

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



*The rare Kurinji flower or Neelakurinji flower
in the Nilgiri hills*

10 The middle portion of the Western Ghats run from 16°N latitude upto Nilgiri hills.

11 This part is made of granites and present rough topography, this area is covered with dense forest.

12 The Nilgiri Hills join the Sahyadris and this marks the junction of the Western Ghats with the Eastern Ghats.

13 Doda Betta is one of the important peak of this area.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



14 The southern part of the Western Ghats is separated from the main Sahyadri range by Pal ghat Gap (Palakkad Gap).

15 Pal ghat Gap is actually a rift valley. This gap is used by a number of roads and railway lines to connect the plains of Tamil Nadu with the coastal plain of Kerala.

16 It is through this gap that moist-bearing clouds of the south-west monsoon can penetrate some distance inland, bringing rain to many areas.

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HILL RANGES OF THE PENINSULAR PLATEAU

4 WESTERN GHATS OR SAHYADRIS



17 Anai Mudi is the highest peak in the whole of southern India.

18 This a nodal point from which the three ranges radiate in different directions.

19 These ranges are the Anaimalai to the north, the Palani to the north-east and the Cardamom Hills to the south.



Tapi River

WESTERN GHATS

▲ NILGIRI HILLS

▲ ANAIMALAI HILLS

▲ CARDAMOM HILLS

KANYAKUMARI

Gulf of Mannar

n Sea

MYANMAR

Bay of Bengal

LAKSHADWEEP
(INDIA)

ANDAMAN & NICOBAR ISLAND
(INDIA)

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HILL RANGES OF THE PENINSULAR PLATEAU

5

EASTERN GHATS



- 1 Eastern Ghats run parallel to the east coast of India leaving broad plains between their base and the coast.
- 2 In striking contrast with continuous eminence of the Western Ghats, it is a chain of highly broken and detached hills starting from the Mahanadi in Odisha to the Vaigai in Tamil Nadu.
- 3 They neither have structural unity nor physiographic continuity. Therefore these hill groups are generally treated as independent units.

Arabian Sea

LAKSHADWEEP
(INDIA)

LAKSHADWEEP SEA

Gulf of
Mannar

Vaigai R.

SHEVAROY
HILLS

JAVADI HILLS

PALKONDA

NALLAMALA

Krishna R.

Godavari R.

MAHENDRAGIRI

Mahanadi R.

Bay of Bengal



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HILL RANGES OF THE PENINSULAR PLATEAU

5

EASTERN GHATS



4

In fact, they almost disappear between the Godavari, and the Krishna.

5

The name Eastern Ghats is therefore, a misnomer since various mountains and hill groups are generally treated as independent units.

6

It is only in the northern part, between the Mahanadi and the Godavari, the Eastern Ghats exhibit true mountain character.

7

The Mahendragiri peak is the tallest peak in this region.

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HILL RANGES OF THE PENINSULAR PLATEAU

5

EASTERN GHATS

8

Between the Godavari and the Krishna rivers, the Eastern Ghats lose their hilly character and they reappear as more or less a continuous hill range in Cuddapah and Kurnool districts of Andhra Pradesh.



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HILL RANGES OF THE PENINSULAR PLATEAU

5

EASTERN GHATS



9

Here, the **Nallamalai Range** with general elevation of 600-850 m is the most prominent feature.

10

The southern part of this range is called the **Palkonda range**.

11

To the south, the hills and plateaus attain very low altitudes, **only Javadi Hills and the Shevroy Hills** form two distinct features of 1,000 m elevation.

12

Further south, the **Eastern Ghats** merge with the **Western Ghats**.

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SIGNIFICANCE OF THE PENINSULAR PLATEAU



Sandstone



Marble

- 1 The Peninsular Plateau of India is the **oldest and mostly a stable landmass** of the Indian sub-continent.
- 2 There are **huge deposits of iron, manganese, copper, bauxite, chromium, mica, gold, etc.**
- 3 Above all **98% of the Gondwana coal deposits of India** are found in the Peninsular Plateau.
- 4 Besides there are **large reserves of shale, sandstone, marble, etc.**

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SIGNIFICANCE OF THE PENINSULAR PLATEAU



Tea and Coffee plantations

5 A large part of north-west plateau is covered with **fertile black lava soil** which is extremely useful for growing cotton.

6 Some other areas of Peninsular plateau are suitable for the cultivation of **plantation crops** like tea, coffee, rubber, etc.

7 Some low lying areas of the plateau are suitable for **growing rice**.

8 A variety of **tropical fruits** is also grown here.

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SIGNIFICANCE OF THE PENINSULAR PLATEAU



Ooty



Kodaikanal

- 9 The highlands of plateaus are covered with different types of forest which provide a large variety of forest products.
- 10 The **rivers** originating in the Western Ghats provide **good** irrigation to the agricultural crops.
- 11 The plateau is also known for its hill resorts such as **Ooty, Kodaikanal, Mahabaleshwar, Mount Abu, etc.**

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